

Shanghai Sieyuan Low Voltage Switchgear Co., Ltd.

SYW6GZ-4000 Switch-disconnector

Product Specifications

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Chapter I Product Overview



1.1 Scope of application

This SYW6GZ-4000 DC switch-disconnector is suitable for DC systems with a rated operating voltage of DC1500V and below and a rated operating current of 4000A and below, providing functions of main circuit connection, breaking, and isolation. Typical application scenarios include energy storage, photovoltaics, and DC distribution.

1.2 Product features

Small volume

The dedicated DC two-pole product has a width of only 259mm, which reduces its volume by up to 40% compared to traditional AC-to-DC conversion products, helping customers save cabinet size.

Low power consumption

Long-term through-flow temperature rise < 50K.

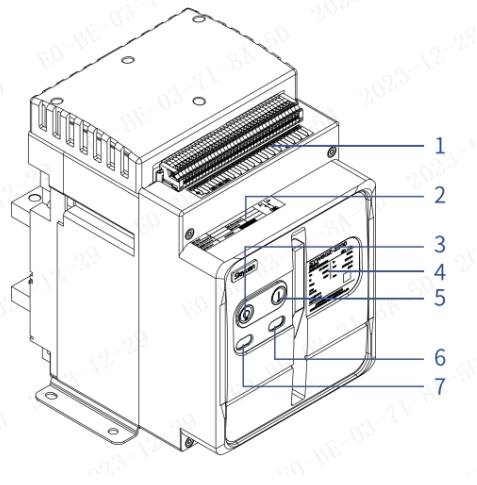
High reliability

The two-pole structure reduces the system force value by 50%, significantly improves the product reliability, and has a mechanical life of up to 10,000 times.

1.3 Standards

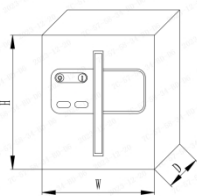
GB/T 14048.3、IEC 60947-3、EN 60947-3、AS 60947-3(TUV-028890-E)

1.4 Structure and indications



- 1. Control circuit wiring instructions
- 2. Specification label
- 3. Opening button
- 4. Nameplate
- 5. Closing button
- 6. Energy release and storage indications
- 7. Opening and closing indications

1.5 Main technical parameters

Model			SYW6GZ-4000
Number of poles （P）			2
Rated current Ie （A）			1600、2000、2500、3200、3600、4000
Rated operational voltage Ue （V）			DC1500
Nominal insulation voltage Ui （V）			DC1500
Rated impulse withstand voltage Uimp （kV）			12
Rated short-circuit current (kA)			100
Short circuit withstand current Icw (kA/s)			150kA/0.2s 100kA/1s
Closing time （ms）			Maximum 70
Utilization category			DC-PV2
Lift cycle (times)	Mechanical	Maintenance-free	10,000
	Electrical	DC1500V	500
	External dimensions Inches H×W×D(mm*mm*mm)		394×259×367
Weight （kg）			40
Attestation			CCC、CE、TUV、AS

Chapter II Scope of Application

2.1 Operating environment

2.1.1 Ambient temperature

Applicable ambient temperature: -40°C to $+70^{\circ}\text{C}$; the average within 24 h shall not be more than $+35^{\circ}\text{C}$.

Storage conditions: The air temperature is $-40^{\circ}\text{C}\sim+80^{\circ}\text{C}$.

Derating not required in high temperature environment.

Table 1

Ambient temperature	$+55^{\circ}\text{C}$	$+60^{\circ}\text{C}$	$+65^{\circ}\text{C}$	$+70^{\circ}\text{C}$
Current compensation constant	1	1	1	1

2.1.2 Atmospheric environmental conditions

The relative air humidity at a maximum temperature of $+40^{\circ}\text{C}$ should not exceed 50%. A higher humidity is allowed at a lower temperature, but special measures should be taken to address occasional condensing due to temperature fluctuation.

2.1.3 Altitude

The applicable altitude for installation is up to 5,000 m, and the insulation and current shall not be derated.

Table 2

Altitude	2000m	3000m	4000m	5000m
Insulation compensation constant	1	1	1	1
Current compensation constant	1	1	1	1

2.2 Corrosion resistance grade

Salt spray: harsh level 2.

2.3 Pollution grade

Pollution level: 3.

2.4 Shockproof requirements

- ◆ The switch-disconnector can ensure resistance to electromagnetic or mechanical shock, and has passed the GB/T 4798.3 standard test;
- ◆ Amplitude : $\pm 1.5\text{mm}$ (2Hz~9Hz);
- ◆ Constant acceleration: 5m/s^2 (9Hz~200Hz) .

2.5 Installation conditions

With the vertical gradient no more than 5° , the switch-disconnector shall be installed in places without explosion danger, conductive dust or the possibility of corroding metal and damaging the insulation.

2.6 Installation category

The installation category of the switch-disconnector's main circuit and power transformer primary coil is IV; the rest auxiliary circuit and control circuit installation category is III.

2.7 Protection grade

Switch-disconnector front IP20, arc extinguishing chamber screen IP10.

Chapter III Accessories

3.1 Closing coil

	Rated voltage (V)	AC220/230	AC380/400	DC110	DC220	DC24
	Instantaneous power (W、VA)	500	550	500	550	420
	Operation voltage (V)	(0.85~1.1) Ue				
	Closing time (ms)	No more than 70ms				

3.2 Shunt trip coil

	Rated voltage (V)	AC220/230	AC380/400	DC110	DC220	DC24	AC220/230 (The holding type)	DC220 (The holding type)
	Instantaneous power (W、VA)	500	550	500	550	420	600	600
	Soak power (W)	/					5	5
	Operation voltage (V)	(0.7~1.1) Ue						
	Break-time (ms)	No more than 30ms						

3.3 Motor operating mechanism

	Rated voltage (V)	AC220/230	AC380/400	DC110	DC220	DC24
	Instantaneous power (W、VA)	820	870	820	880	810
	Steady-state power (W)	215	230	215	270	260
	Operation voltage (V)	(0.85~1.1) Ue				

3.4 Auxiliary contact

	Rated heating current (A)	6
	Rated insulation voltage Ui (V)	660
	Capacity	AC-15: 380V 0.95A DC-13: 220V 0.63A

3.5 Undervoltage trip coil

	Rated voltage (V)	AC220/230	AC380/400	DC110	DC220	DC24
	Instantaneous power (VA)	350	520	620	350	400
	Service rating (W)	5.5	7.3	3.1	5.5	3.5
	Operation voltage (V)	$(0.35 \sim 0.7) U_e$				
	Reliable switching voltage	$(0.85 \sim 1.1) U_e$				
	Reliable uncloseable voltage	$\leq 0.35 U_e$				
	Delay time	instantaneous、1s、3s、5s				

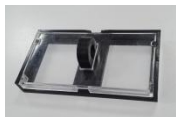
3.6 No-voltage trip coil

	Rated voltage (V)	AC220/230	AC380/400
	Instantaneous power (VA)	135	330
	Service rating (W)	3.7	6.4
	Operation voltage (V)	$(0 \sim 0.65) U_e$	
	Reliable switching voltage	$(0.85 \sim 1.1) U_e$	
	Reliable uncloseable voltage	$\leq 0.35 U_e$	
	Delay time	instantaneous、1s、3s、5s	

3.7 Split position lock

	It locks the disconnecting switch in the opening position to ensure that it cannot be closed
	One lock and one key: One disconnecting switch with one lock and one key Two locks and one key: Two disconnecting switches with two locks and one key Three locks and one key: Three disconnecting switches with three locks and one key Three locks and two keys: Three disconnecting switches with three locks and two keys Five locks and three keys: Five disconnecting switches with five locks and three keys

3.8 Button locked



The opening and closing buttons on the mask can be locked to prevent misoperation.

(Note: The padlock shall be provided by the customer)

3.9 Door case



It is installed on the door of the cabinet chamber for sealing, making the protection level of the disconnecting switch reach IP40.

3.10 Fuse



Rated current	Hollyfuse
3300A	H5FG3300
3500A	H5FG3500
3800A	H5FG3800
4000A	H5FG4000

Note 1: The reduction coefficient of the fuse is detailed in the specification of the corresponding manufacturer. The recommended reduction coefficient range is 0.6~0.8;

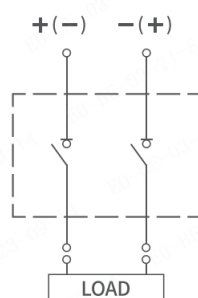
Note 2: If you choose other fuses, please confirm the fuse parameters with us.

Chapter IV Wiring Mode, Outline, and Installation Dimensions

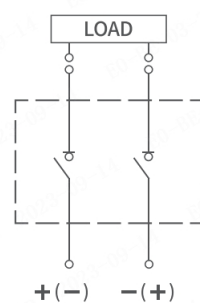
4.1 Mode of connection

4.1.1 Connection mode of the switch-disconnector

Upper incoming line

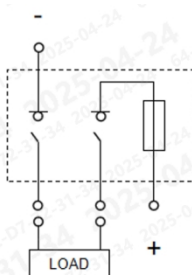


Lower incoming line

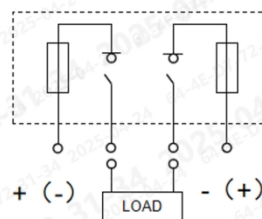


4.1.2 Select the fuse wiring mode for the switch-disconnector

With Single-sided fuse

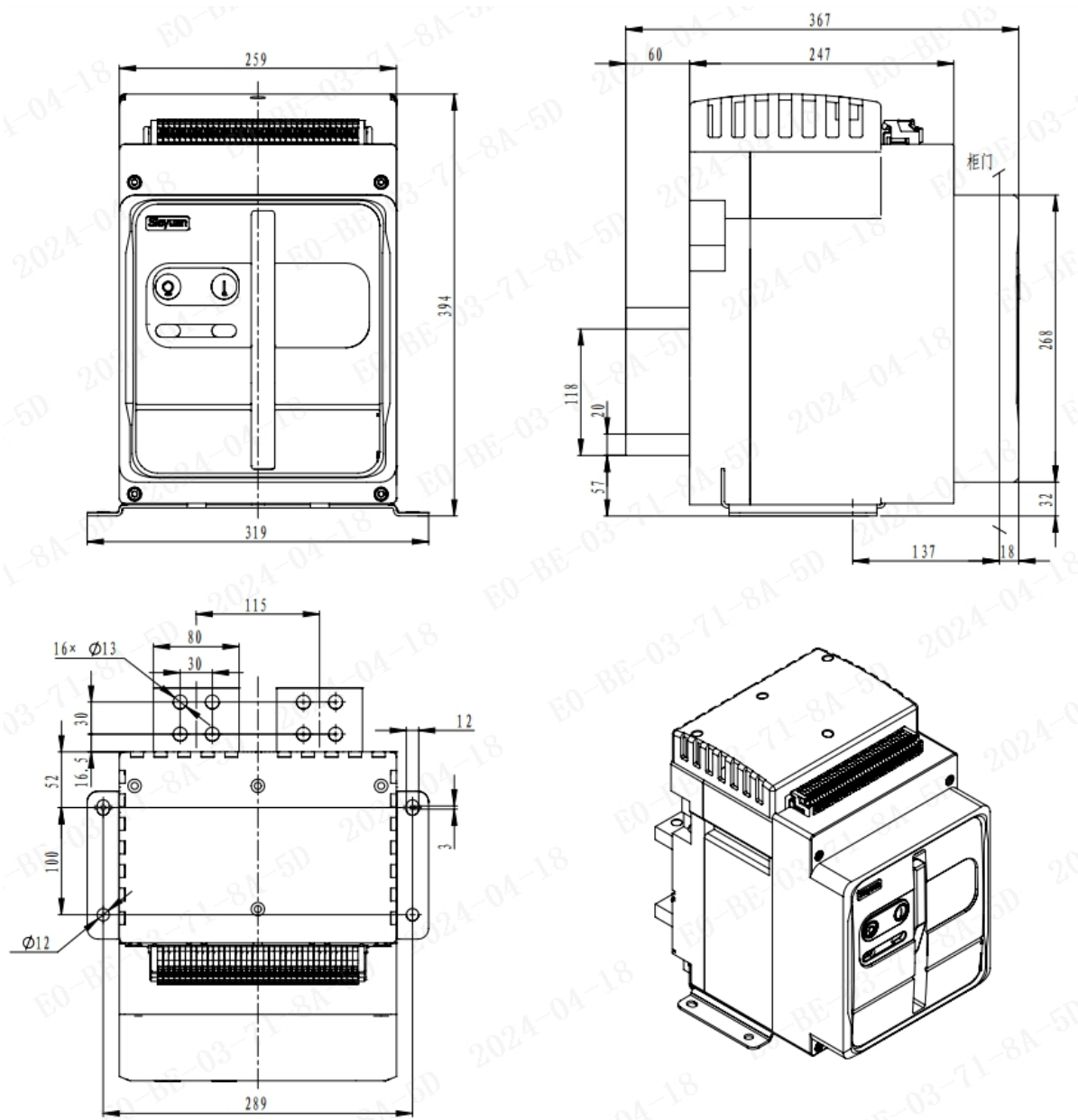


With two-sided fuse



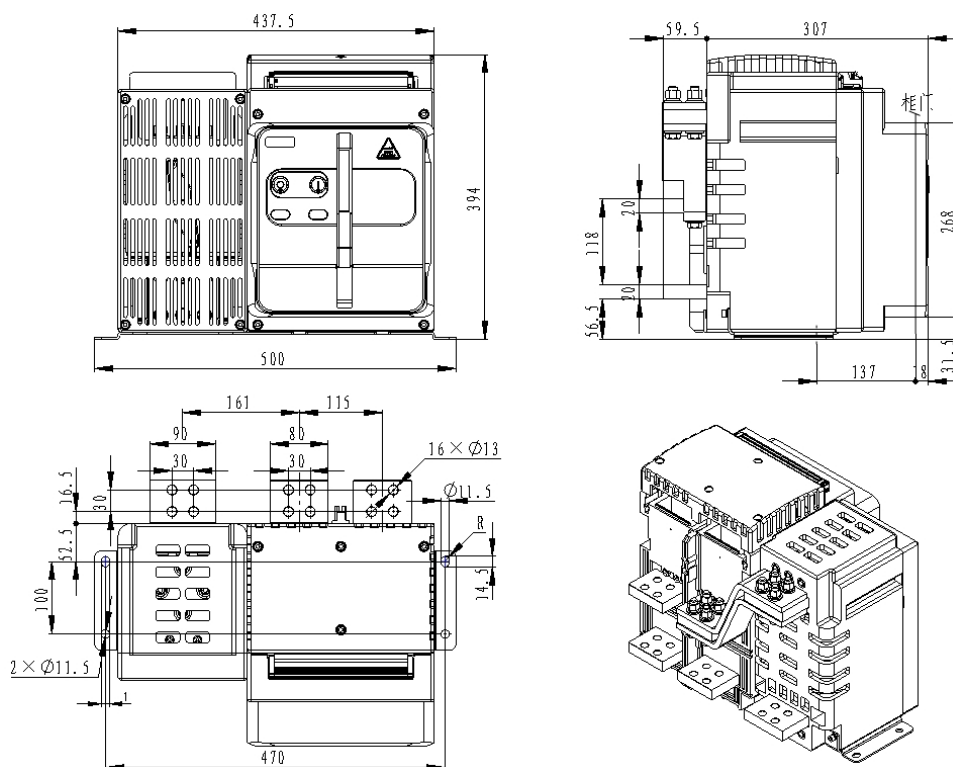
4.2 Outline overall and installing dimensions

4.2.1 DC switch-disconnector



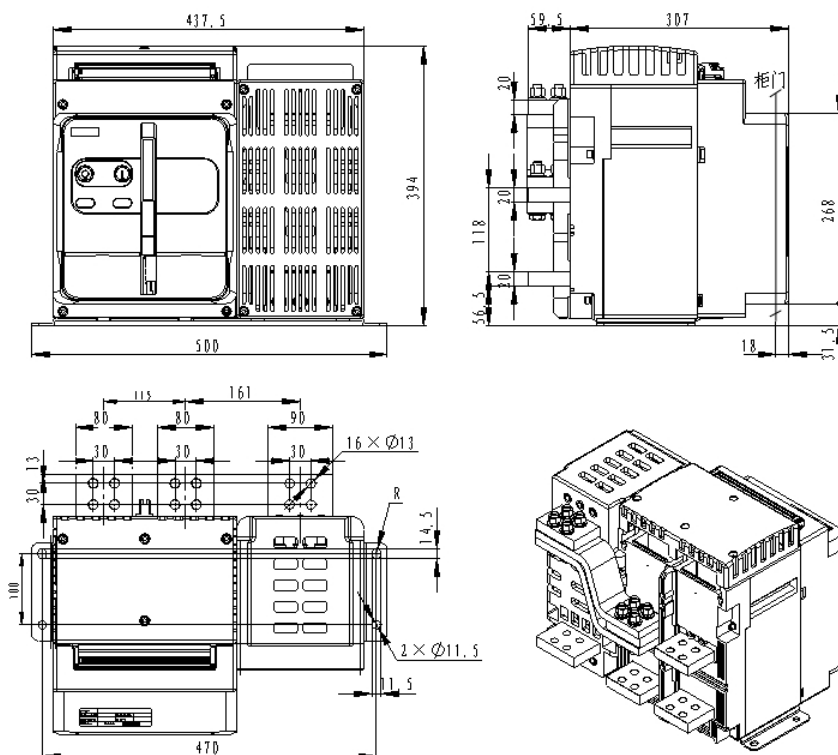
Note: The busbar is connected by M12 bolts of Grade 8.8 and contact gaskets at tightening torque of 60 N·m.

4.2.2 With single-sided fuse (Fuse is configured on the left to adapt combiner cabinet)



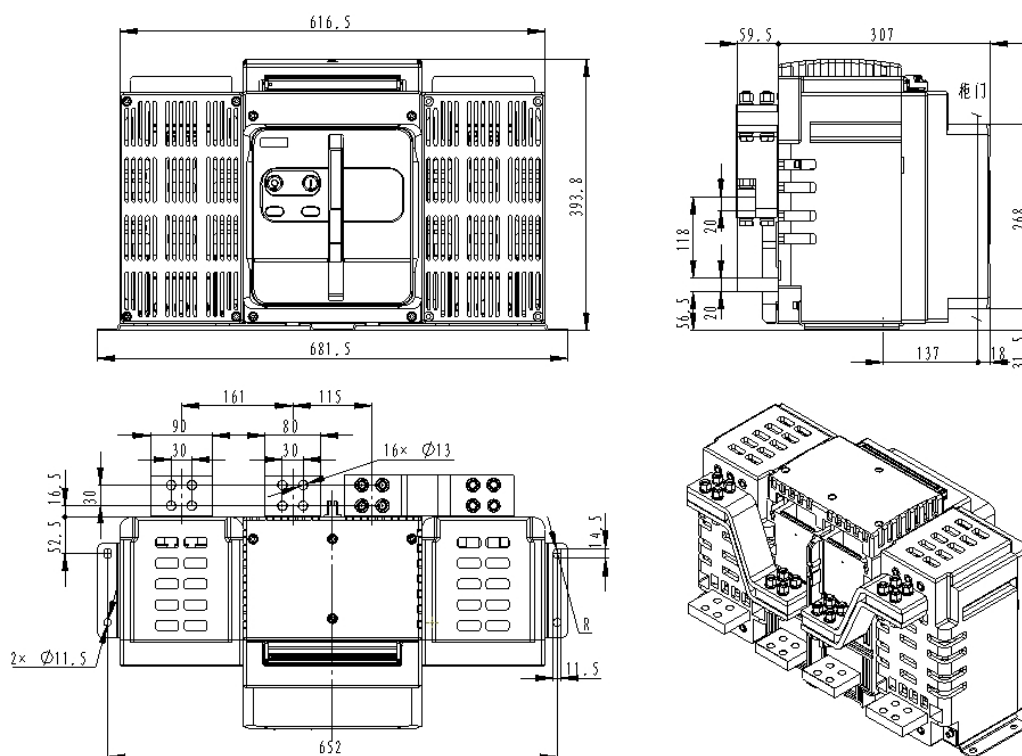
Note: The busbar is connected by M12 bolts of Grade 8.8 and contact gaskets at tightening torque of 60 N·m.

4.2.3 With single-sided fuse(Fuse is configured on the right to adapt combiner cabinet)



Note: The busbar is connected by M12 bolts of Grade 8.8 and contact gaskets at tightening torque of 60 N·m.

4.2.4 With two-sided fuse (Two-sided configuration of fuses to adapt combiner cabinet)



Note: The busbar is connected by M12 bolts of Grade 8.8 and contact gaskets at tightening torque of 60 N·m.

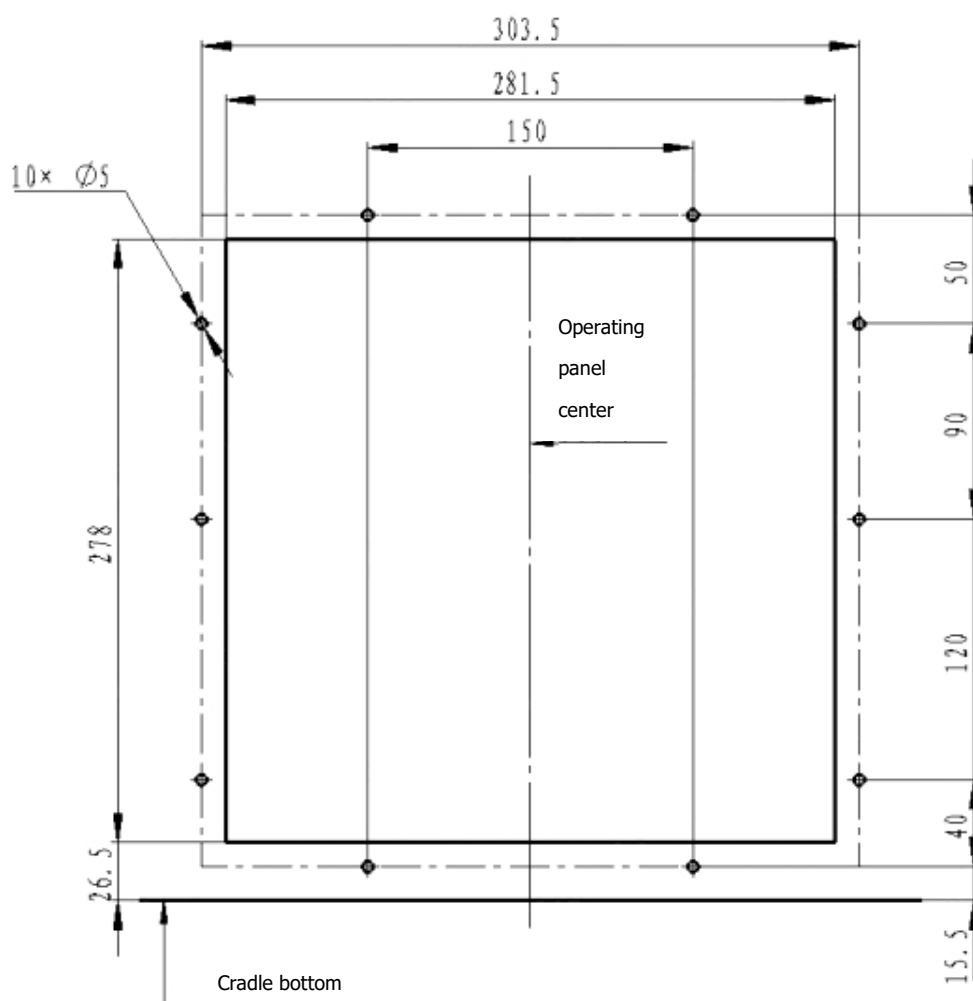
4.3 Reference Specifications of Switch-disconnector's Main Circuit Copper Bar

Rated current of frame	Rated current (A)	Copper bar specifications	
		Radical	size (mm)
SYW6GZ-4000	1600	2	100×5
	2000	3	100×5
	2500	4	100×5
	3200	5	100×10
	3600	5	100×10
	4000	5	100×10

Note: The customer shall design the connection copper bar specification of the fuse selected for the switch-disconnector according to the actual situation.

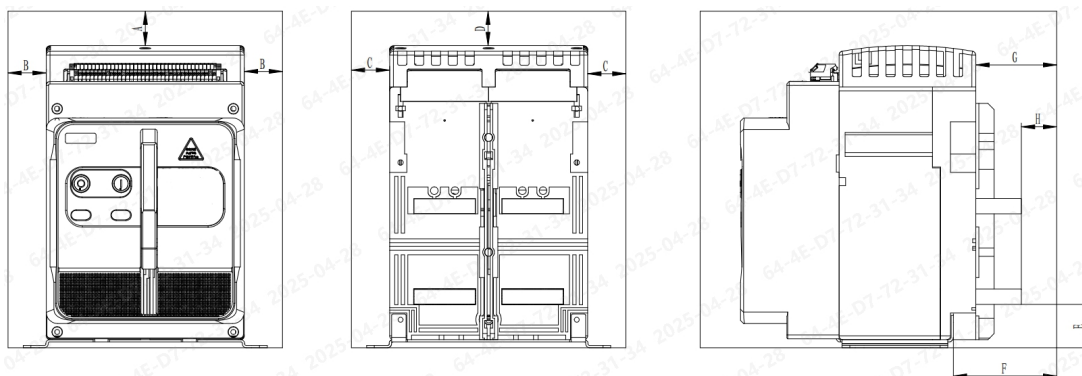
4.4 Panel front door

Mounting hole(unit: mm)

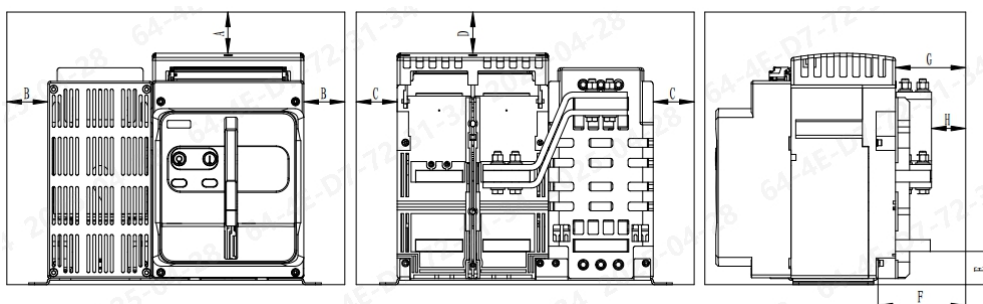


4.5 Safe Distance (unit: mm)

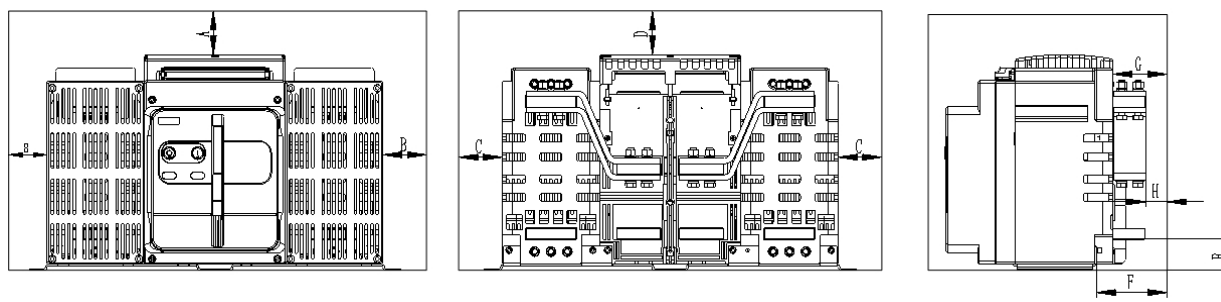
Switch -disconnectors:



The switch-disconnector is equipped with a fuse (single side fuse):



The switch-disconnector is equipped with a fuse (two-sided fuse):



	To the insulator ($\geq$)	To the metallic body ($\geq$)
A	0	60
B	60	60
C	60	60
D	0	60
E	0	60
F	0	60
G	60	60
H	0	60

Note: If the distance to the live body does not meet the requirements in the table, insulation measures should be added between the busbar and the live body.

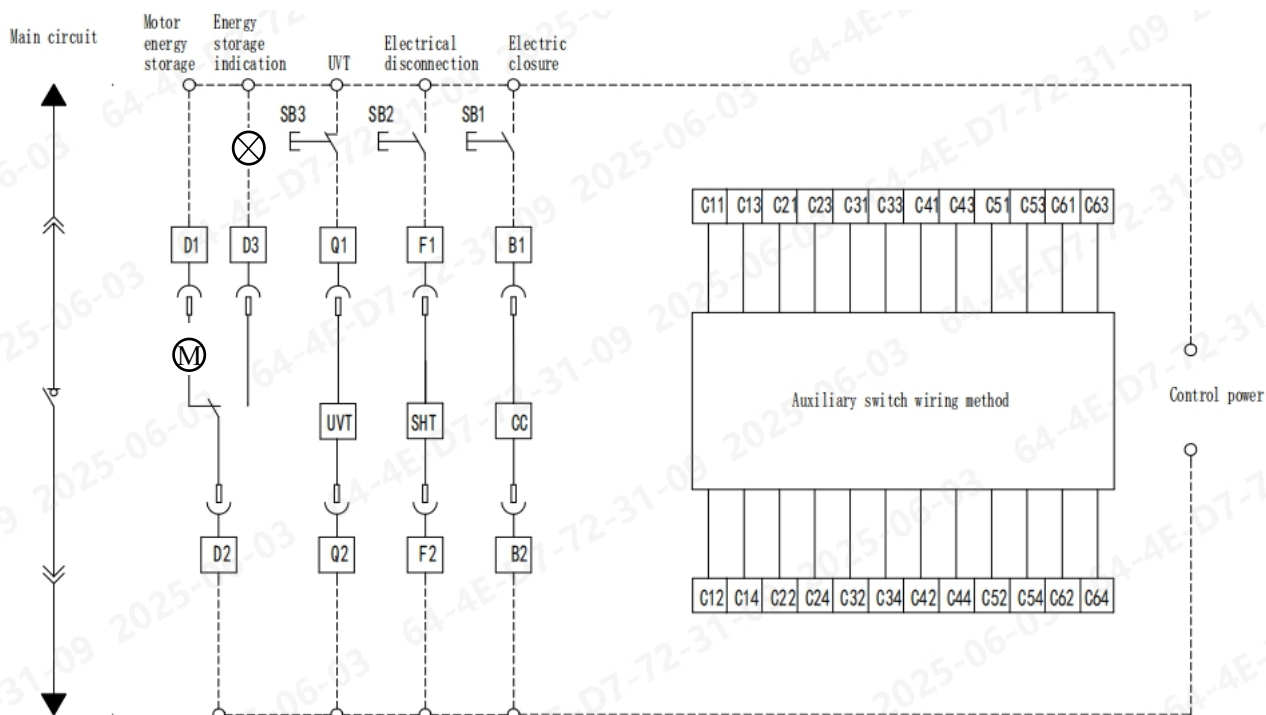
4.6 Power loss (unit: W)

Rated current of frame	Power consumption (single pole)
SYW6GZ-4000	≤ 150

Note: The above power loss of the switch-disconnector is measured at the laboratory ambient temperature $+40^{\circ}\text{C}$.

Chapter V Secondary Schematic Diagram

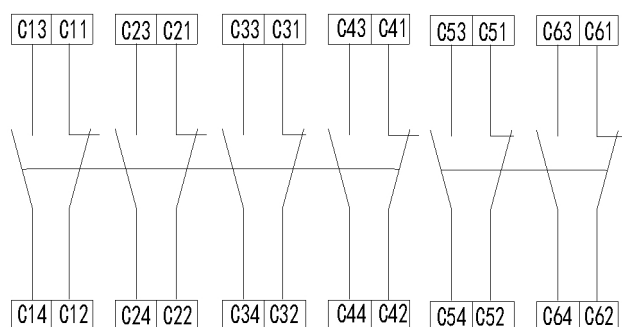
5.1 Secondary schematic diagram of the switch-disconnector



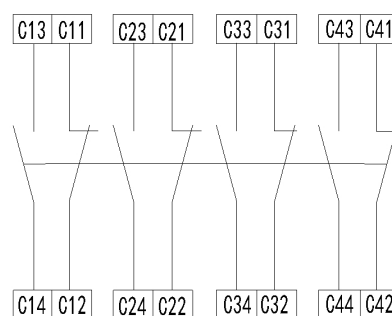
SB1 — closing button (provided by the customer); SB2 — shunt button (provided by the customer);
 SB3 — undervoltage disconnection button (provided by the customer)

Note: The current state of the switch-disconnector is de-energized, open, and without energy storage;

Six Open and Six closed



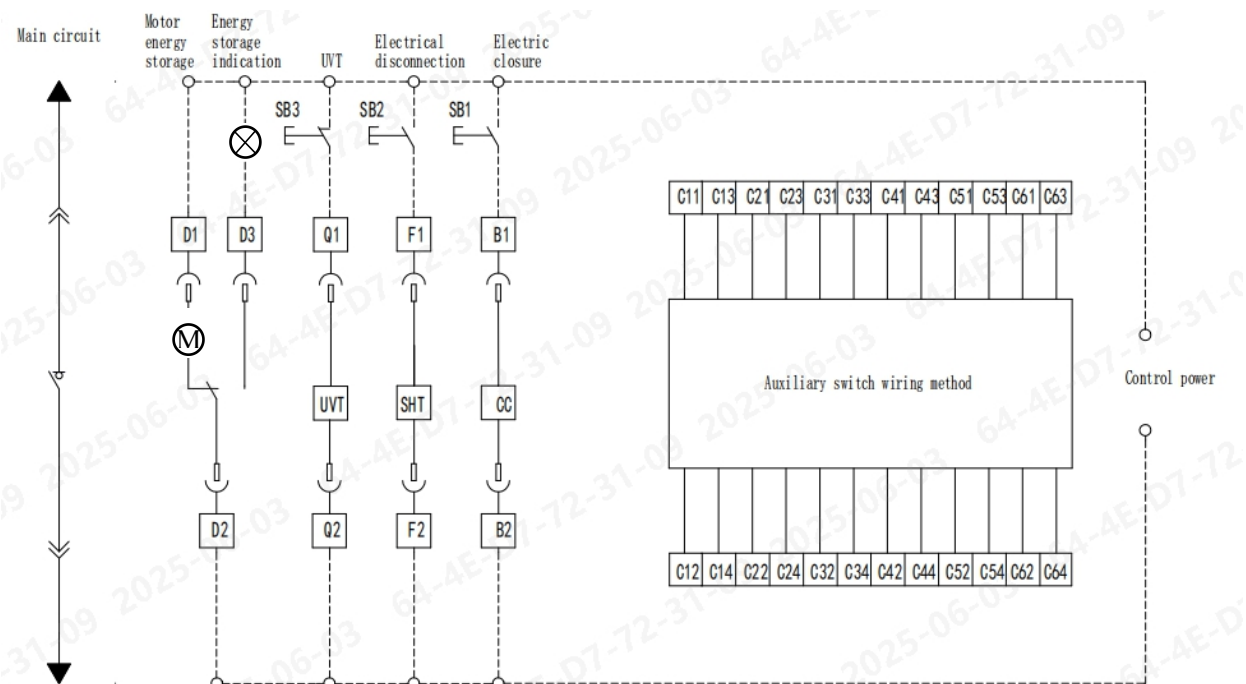
Four Open and Four Closed



5.2 Secondary schematic diagram of the switch-disconnector with fuse

The secondary electrical diagram for switch-disconnector equipped with fuses consists of the electrical accessories of the switch-disconnector (such as motors, undervoltage trip units, de-energizing trip units, closing electromagnets, and auxiliary contacts) and the electrical accessories of the fuse (including the fuse indicator). Depending on the selection of the auxiliary contacts of the switch-disconnector and the electrical indicator of the fuse (as specified in the order specifications), there are three forms of the secondary electrical wiring diagram.

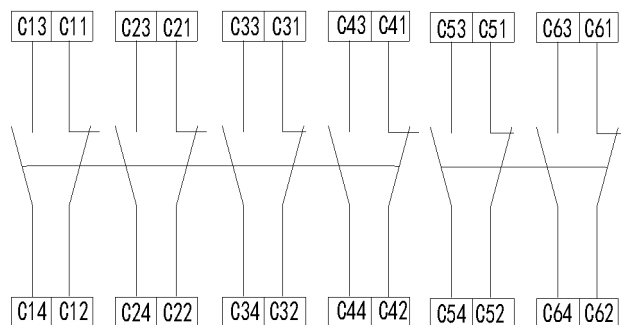
Form 1: Switch-disconnector auxiliary contact selection: four open and four closed / six open and six closed; fuse electrical indication selection: no;



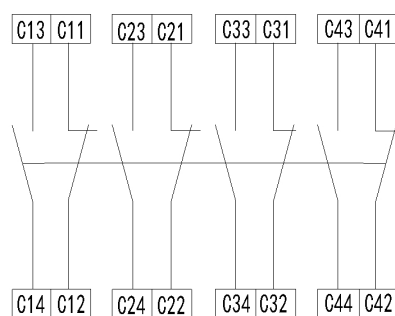
SB1 — closing button (provided by the customer); SB2 — shunt button (provided by the customer);
SB3 — undervoltage disconnection button (provided by the customer)

Note: The current state of the switch-disconnector is de-energized, open, and without energy storage;

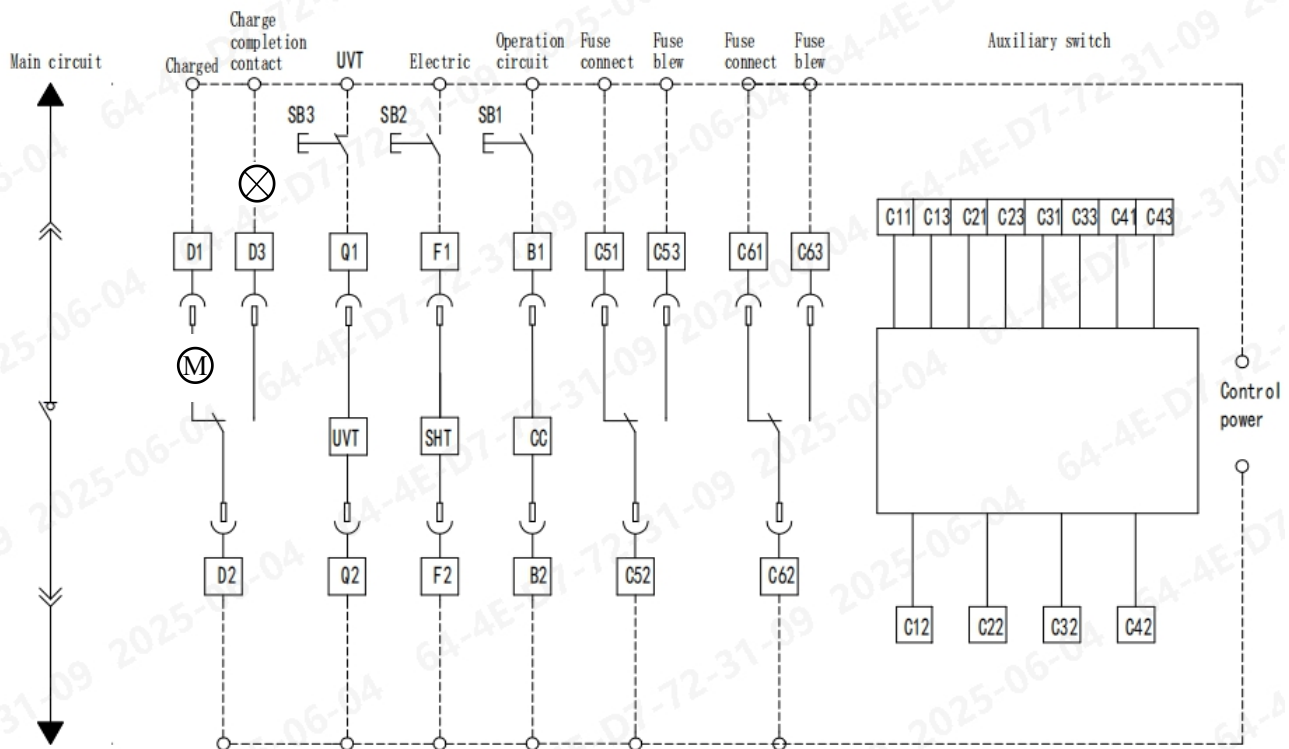
Six open and six closed



Four open and four closed



**Form 2: Selection of auxiliary contacts for switch-disconnector: four open and four closed;
selection of electrical indicators for fuses: need.**

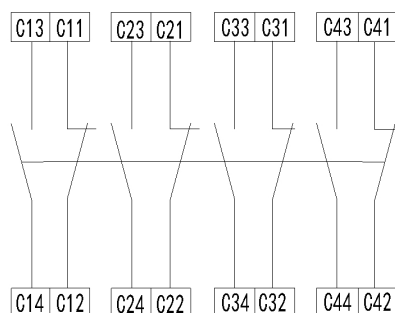


SB1 — closing button (provided by the customer); SB2 — shunt button (provided by the customer);
SB3 — undervoltage disconnection button (provided by the customer)

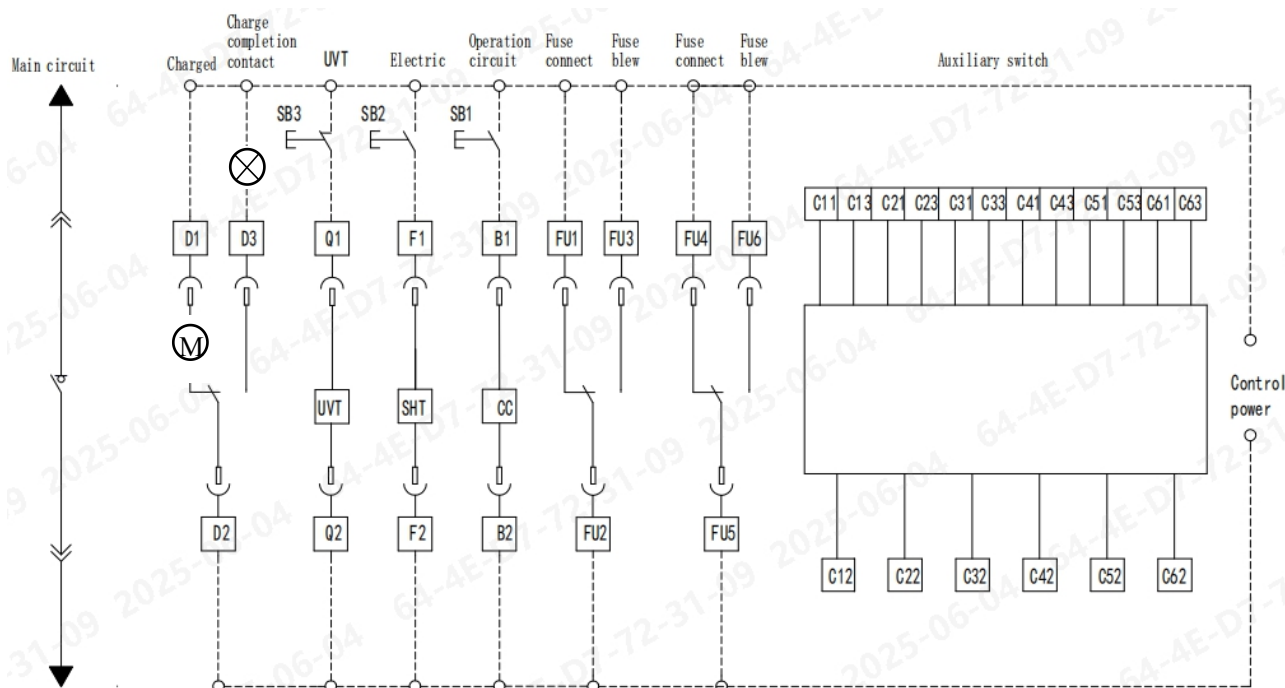
Note:

1. The current state of the switch-disconnector with fuse is not energized, open and unenergized;
2. When the fuse configuration is selected as single side, the electrical indication of the fuse is only connected to C51, C52, C53, and C61, C62, C63 are not connected.

Four Open and Four Closed



**Form 3: Selection of auxiliary contacts for switch-disconnector: six open and six closed;
selection of electrical indicators for fuses: need.**

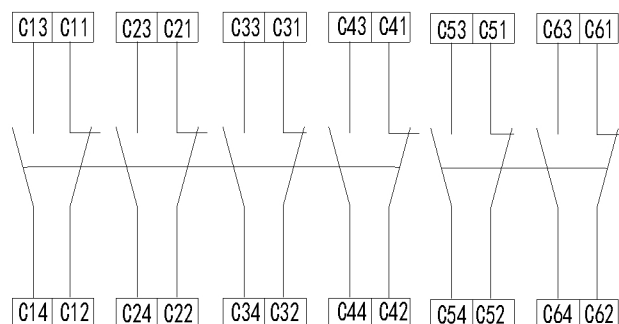


SB1 — closing button (provided by the customer); SB2 — shunt button (provided by the customer);
SB3 — undervoltage disconnection button (provided by the customer)

Note:

1. The current state of the switch-disconnector with fuse is not energized, open, and unenergized;
2. FU1, FU2, FU3, FU4, FU5, FU6 are external signal lines, which are not integrated into the product body, and the line length can be customized;
3. When the fuse configuration is selected as single side, the electrical indication of the fuse is only connected to FU1, FU2, FU3, and FU4, FU5, FU6 are not connected.

Six Open and Six Closed



Model Explanation and Code Specification

SY W 6 GZ - □ □ □ / □ / □ / □ / □ / □ / □ / □ / □ / □ / □ / □ / □
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

S/N	Name	Specification and type code	Description
1	Enterprise code	SY — Sieyuan	
2	Product type code	W — Universal	
3	Design S/N	6	
4	Series derivation code	GZ — Isolated DC	
5	Case frame current	40-4000A	
6	Special-purpose product code	Not marked — Standard	
7	Breaking capacity class	Not marked — Standard	
8	Rated current	16-1600A、20-2000A、25-2500A、32-3200A、36-3600A、40-4000A	
9	Rated voltage	15-1500V	
10	Number of poles	2-2P	
11	Installation method	G — Fixed	
12	Wiring mode	1 — Horizontal wiring	
13	Shunt trip coil	F1-AC380/400V、F2-AC220/230V、F3-DC220V、F4-DC110V、F6-DC24V、 F7-AC220/230V（holding type）、F8-DC220V（holding type）	
	Closing coil	B1-AC380/400V、B2-AC220/230V、B3-DC220V、B4-DC110V、B6-DC24V	
	Motor operating mechanism	D1-AC380/400V、D2-AC220/230V、D3-DC220V、D4-DC110V、D6-DC24V	
	Auxiliary switch	FC3-Four open and four closed、FC4-Six open and six closed	
14	Undervoltage/No-voltage trip coil	Q1-AC380/400V、Q2-AC220/230V、Q3-DC220V、Q4-DC110V、Q6-DC24V、S1-AC380/400V、S2-AC220/230V	Choose either undervoltage trip coil or No-voltage trip coil, omitted if there is no such accessory
	Delay time of undervoltage release	Undervoltage: 0 — instantaneous, 1 — 1 s delay, 3 — 3 s delay, 5 — 5 s delay	
		No-voltage: 0 — instantaneous, 1 — 1 s delay, 3 — 3 s delay, 5 — 5 s delay	
	Button blocking	AS — Button blocking	Omitted if there is no such accessory
	Door frame	MK — Door frame	Omitted if there is no such accessory
	Opening position lock	FS11 — one lock and one key, FS21 — two locks and one key, FS31 — three locks and one key, FS32 — three locks	Choose one from five,

		and two keys, FS53 — five locks and three keys	omitted if there is no such accessory
15	Identification code	Not marked — Standard	
16	Special customization code	Customer specific requirements	
17	Fuses are configured	01-One side (Left side), 02-One side (Right side) 03-Both sides	
	Transparent protective cover	TZ-Transparent protective cover	
18	Fuses manufacturer	HL-Hollyfuse	
	Rated current of fuse	33-3300A、35-3500A、38-3800A、40-4000A	
	Electrical indications	WG- inches switch	
	Circuit breaker breaking capacity	15-150kA、25-250kA、□ 30-Other	

(Please fill in the number on _____, and tick $\checkmark$ in $\square$. See the instructions for other content.)

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	Delay time of undervoltage/No-voltage trip coil	☐ 0 — instantaneous, ☐ 1 — 1 s delay, ☐ 3 — 3 s delay, ☐ 5 — 5 s delay	Please do not tick if there is no such accessory
	Button blocking	☐ AS — Button blocking	
	Door frame	☐ MK — Door frame	
	Opening position lock	☐ FS11 — one lock and one key, ☐ FS21 — two locks and one key, ☐ FS31 — three locks and one key, ☐ FS32 — three locks and two keys, ☐ FS53 — five locks and three keys	
	Interphase barrier	☒ Interphase barrier	Standard configuration
15	Identification Code	Not marked-Standard	
16	Special customization code	Not marked-Standard	
17	Fuse configuration	☐ 01-One side (Left side)、 ☐ 02-One side (Right side)、 ☐ 03-Both sides	
	Transparent protective cover	☐ TZ — transparent protective cover	
18	Fuses manufacturer	☐ HL-Hollyfuse H5FG	
	Fuses rated current	☐ 33-3300A、 ☐ 35-3500A、 ☐ 38-3800A、 ☐ 40-4000A	
	Electrical indications	☐ WG-Fuse indication (When the auxiliary contact is selected as six open and six closed, the electrical indication is not integrated into the frame body, and the electrical indication signal line is an external line, and the length of the line can be customized according to the requirements)	
	Circuit breaker breaking capacity	☐ 15-150kA、 ☐ 25-250kA、 ☐ 30-Other	
Note: If there are any special requirements, please explain them in the column of "Special requirements".			
Model and specifications	SYW6GZ-___/___/___/___/___+___+___+___/___+___+___+___/___+___/___+___+___		